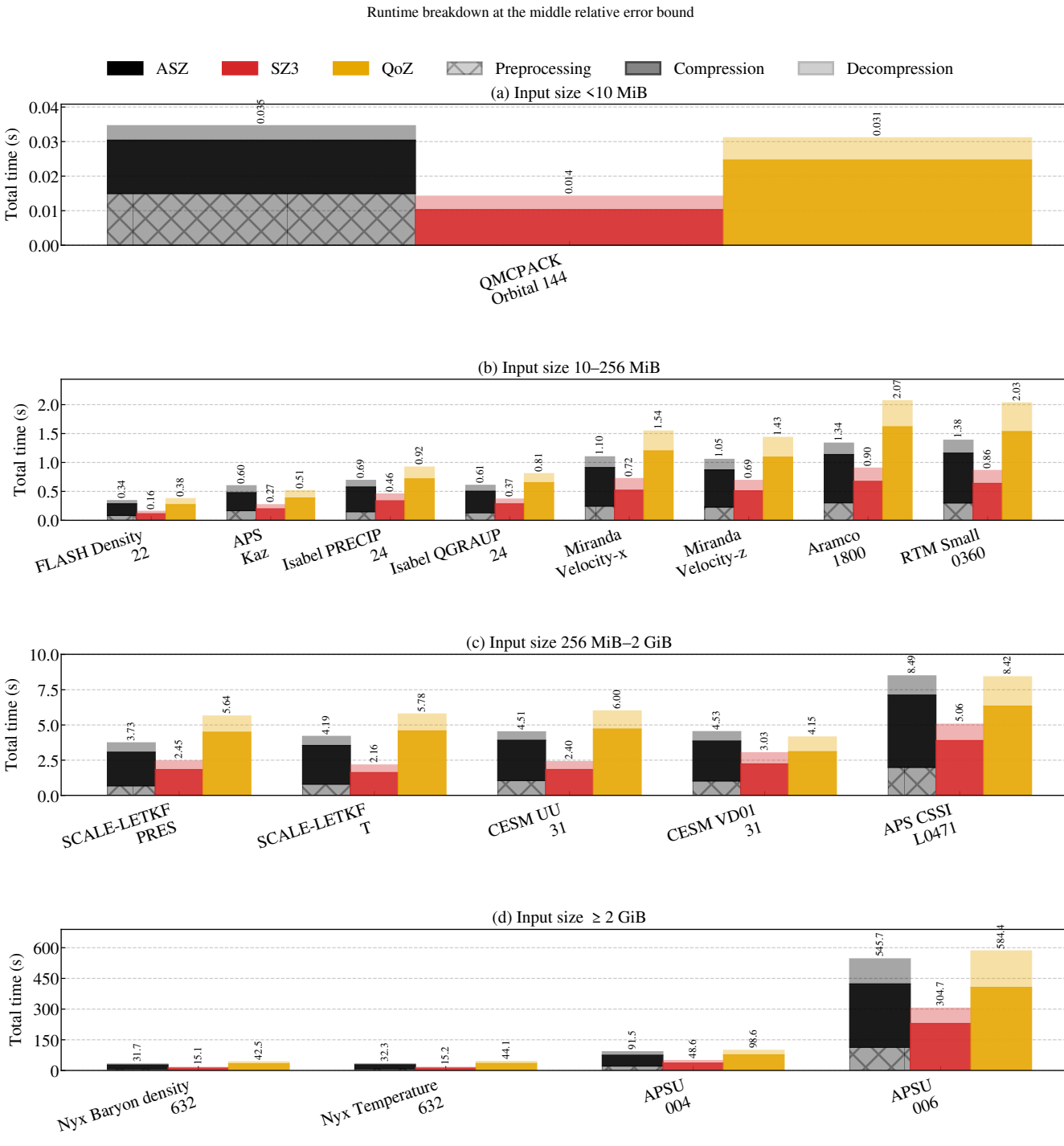
Runtime breakdown at the middle relative error bound
ASZ
SZ3
QoZ
Preprocessing
Compression
Decompression
(a) Input size <10 MiB
Total time (s)
0.04
0.03
0.02
0.01
0.00
0.035
0.014
0.031
QMCPACK
Orbital 144
(b) Input size 10–256 MiB
Total time (s)
2.0
1.5
1.0
0.5
0.0
0.34
0.16
0.38
0.60
0.27
0.51
0.69
0.46
0.92
0.61
0.37
0.81
1.10
0.72
1.54
1.05
0.69
1.43
1.34
0.90
2.07
1.38
0.86
2.03
FLASH Density
22
APS
Kaz
Isabel PRECIP
24
Isabel QGRAUP
24
Miranda
Velocity-x
Miranda
Velocity-z
Aramco
1800
RTM Small
0360
(c) Input size 256 MiB–2 GiB
Total time (s)
10.0
7.5
5.0
2.5
0.0
3.73
2.45
5.64
4.19
2.16
5.78
4.51
2.40
6.00
4.53
3.03
4.15
8.49
5.06
8.42
SCALE-LETKF
PRES
SCALE-LETKF
T
CESM UU
31
CESM VD01
31
APS CSSI
L0471
(d) Input size ≥ 2 GiB
Total time (s)
600
450
300
150
0
31.7
15.1
42.5
32.3
15.2
44.1
91.5
48.6
98.6
545.7
304.7
584.4
Nyx Baryon density
632
Nyx Temperature
632
APSU
004
APSU
006